Andres Enrique Quintero Santander

Department of Mathematical Sciences · Indiana University Indianapolis
Email: aequinte@iu.edu · Arxiv Profile: arXiv Profile · Citizenship: Colombia

1 Research Interests

Dynamical Systems, Topological Dynamics, Complex Dynamics (Hénon maps), and Birational Dynamics.

2 Education

- **Ph.D. in Mathematics** Expected May 2027
  Indiana University Indianapolis, Indianapolis, IN
  Advisors: Prof. Roland Roeder
  Dissertation Title: *[Your Dissertation Title]*

Peer-Reviewed Publications

1. **Andres E. Quintero Santander** and Carlos Uzcátegui. "On the Ellis semigroup of a cascade on a compact metric countable space." *Semigroup Forum*, vol. 101(2), pp. 435–451, 2020.

Preprints and Submitted Papers

2. **Andres E. Quintero Santander**, Roland Roeder, Andrei Grigorev, and Krishna Kalidindi. "Complex dynamics perspective for birational maps of the plane arising from cluster algebra mutations." *Submitted to Arnold Mathematical Journal*. arXiv:2602.02324 [math.DS].

Refereed Conference Proceedings & Extended Abstracts

3. **Andres E. Quintero Santander**. "On the holomorphic and random dynamics for some examples of higher rank free groups generated by Hénon type maps." *Proceedings of the AMATES-ICMAM Workshop for Central America and the Caribbean 2025*, Part of the book series: Trends in Mathematics (TM, volume 22). (Extended Abstract, Solo Author).

Other Research & Interdisciplinary Publications

4. Leidy Portilla, Tulia Rivera and **Andres E. Quintero Santander**. "Estrategias de juego: el caso del baloto." *Proceedings of the International Conference on Statistics: Encuentro de Educación Estócastica*, 2014. (Extended Abstract).

Awards, Fellowships, & Scholarships

- **Graduate Research/Tuition Scholarship**, IU Indianapolis 2026 – 2027
- **Graduate Assistantship/Stipend Award**, IU Indianapolis 2021 – 2026

3 Teaching Experience

3.1 Instructor of Record (Main Lecturer)

Institution: Indiana University Indianapolis (IU Indy) **Fall 2023 – Present**
Department of Mathematical Sciences

- Designed course syllabi, delivered lectures, and managed primary instruction for graduate and undergraduate mathematics courses.
- **Courses Taught:**
  - **MATH I563: Advanced Geometry** (Summer 2026) – *Graduate / Master's-level Course*
  - **MATH M118: Finite Mathematics** (Spring 2026)
  - **MATH I1710: Multidimensional Mathematics** (Fall 2025) – *Taught through the Purdue–IU Consortium to Purdue students*
  - **MATH I231 & I232: Calculus for the Life Sciences I & II** (Fall 2024, Summer 2025)
  - **MATH 16600: Analytic Geometry & Calculus II** (Fall 2023)

Institution: Universidad Industrial de Santander (UIS) **Aug 2018 – Aug 2021**
Adjunct Faculty (Profesor Catedrático) – School of Mathematics at Bucaramanga & School of Mathematics at Barrancabermeja

- Coordinated and led primary undergraduate lectures, created evaluations, and assigned final grades across core departmental subjects.
- **Courses Taught:** Linear Algebra I & II, Calculus I, II & III, Differential Equations, and General Mathematics.

3.2 Teaching Assistant & Recitation Leader

Indiana University Indianapolis (IU Indy) **Fall 2022**

- **MATH 16500: Analytic Geometry & Calculus I**: Led problem-solving recitations, held office hours, and graded assignments for multiple sections (29512, 24553, 24242).

Universidad Industrial de Santander (UIS) **Nov 2016 – Aug 2018; May 2020**

- Assisted main professors with coursework grading and leading student problem-solving sessions.
- **Recitation Leader / Academic Support (Monitorías SEA):** Facilitated small-group math recitations and tutoring sessions via the CREE program (2016–2017, 2020).
- **TA Support Courses:** Calculus I & III, Linear Algebra I, Differential Equations.

Selected Talks and Presentations

- **Contributed Talks**,
  - "On Random walks and Stationary measures for groups generated by Hénon type maps," *AMS Sectional Meeting.*

  - **Invited Seminar Talk:** "Local minimal actions of non-elementary groups and pseudo-groups generated by Hénon Maps," *Dynamics Seminar* Department of Mathematics, University of Notre Dame, 2026.
  - **Invited Technical Seminar:** "Una breve introducción al caos, la entropía y los sistemas dinámicos," *Programa Institucional de Formación y Actualización Tecnológica*, Corporación para la Investigación de la Corrosión (CIC), Colombia, June 26, 2026.
  - **Invited Seminar Talk:** "Sobre el cálculo del grado dinámico para una familia de mapas biracionales inspirados por la teoría de Cluster álgebras," *Topology Seminar "Rafael Isaacs"* (Sesión 463), Escuela de Matemáticas, Universidad Industrial de Santander (UIS), Bucaramanga, Colombia, June 9, 2026.

- **Seminar Talk**, "[Title of your talk or related project]," Dynamical Systems Seminar, Indiana University Indianapolis, Indianapolis, IN, [Month Year].

- **Poster Presentation**,"On the Holomorphic and Random Dynamics For some examples of higher rank Free Groups generated by Hénon-type maps," *Midwest Dynamical Systems Conference*, University of Notre Dame, Notre Dame, IN, May 2–3, 2026.

4 Conferences & Workshops Attended

- **Midwest Dynamical Systems Conference**, University of Notre Dame, Notre Dame, IN, May 2–3, 2026.

- **Midwestern Workshop on Asymptotic Analysis**, Purdue University Fort Wayne, Fort Wayne, IN, October 10–12, 2025.

- **Midwest Dynamical Systems Early Career Conference**, The Ohio State University, Columbus, OH, May 14–15, 2024.

- **Midwest Dynamical Systems Conference**, University of Illinois Chicago, Chicago, IL, October 20–22, 2023.

- **MINT Graduate School: Facets of Complex Dynamics (Summer School)**, Institut de Mathématiques de Toulouse (IMT), Toulouse, France, June 26–30, 2023.

- **Midwestern Workshop on Asymptotic Analysis**, Purdue University Fort Wayne, Fort Wayne, IN, October 7–9, 2022.

- **Midwest Dynamical Systems Conference**, Northwestern University, Evanston, IL, November 12–14, 2021.

Academic Service & Leadership

- **AMS Student Chapter at IU Indianapolis**
  - **Secretary**, Sep 2025 – Aug 2026
  - **Vice President**, Mar 2023 – Aug 2024

References

- **Prof. Roland Roeder** (Advisor/Research Reference)
  Department of Mathematical Sciences, IU Indianapolis · roederr@iu.edu
- **Prof. Julio Rebelo** (Research Reference)
  Institut de Mathématiques de Toulouse at the Université de Toulouse in France. · rebelo@math.univ-toulouse.fr
- **Prof. Jeffrey Diller** (Research Reference)
  Department of Mathematics at the University of Notre Dame in the USA. · jdiller@nd.edu
- **Prof. Carlos Uzcátegui** (Research Reference)
  Department of Mathematics, universidad Industrial de Santander · cuzcatea@saber.uis.edu.co